Leveraging Large Language Models for Reward Shaping in Simulated Swarm Robotics

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Abstract

this paper focus on solving the robot from simulatied environment to real world enviroment. We propose a novel approach that leverages large language models (LLMs) for reward shaping in reinforcement learning (RL) to enhance the performance of swarm robotics in real-world scenarios. Our method involves using LLMs to generate informative and context-aware reward signals based on the collective behavior of the swarm, enabling more efficient learning and adaptation to dynamic environments. We evaluate our approach in a series of simulated swarm robotics tasks, demonstrating significant improvements in learning speed and task performance compared to traditional reward shaping techniques. Furthermore, we validate the effectiveness of our method through real-world experiments with physical robots, showcasing its potential for practical applications in swarm robotics. Our results highlight the promise of integrating LLMs with RL for advancing the capabilities of swarm robotic systems.

1 Core Concepts and Paper Checklist

1.1 Modeling in Swarm Robotics

Modeling in swarm robotics is a complex task due to the dynamic and decentralized nature of these systems. We use Nvidia Isaac Lab to build high-fidelity simulations that replicate real-world swarm robotic environments. These simulations allow us to test and refine our reinforcement learning algorithms in a controlled setting before deploying them on physical robots. By leveraging the capabilities of Nvidia Isaac Lab, we can model interactions between agents, environmental dynamics, and sensor noise, ensuring that our approach is robust and adaptable to real-world scenarios.

1.2 Isacc Lab

NVIDIA Isaac Lab — the robotics simulation and training platform that NVIDIA released (successor to Isaac Gym, built on top of Isaac Sim / Omniverse). A robotics RL training framework on top of Isaac Sim (Omniverse).

Provides high-performance physics simulation (via PhysX) and GPU acceleration.

Designed for robotics researchers working on:

Reinforcement learning (RL)

Imitation learning

Robot control policies

Multi-agent / swarm robotics

Submitted to 39th Conference on Neural Information Processing Systems (NeurIPS 2025). Do not distribute.

1.3 generate basic control for robot

the paper with your own.